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End Semester Examination 2025

B.Tech.: Civil Engineering

Semester: 3rd

Paper Name: Engineering Mathematics-III

Paper Code: TMA 302

Time: 3 Hours

Max. Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) The total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q 1

20 Marks [CO1]

- (a) Find the Fourier series expansion for the periodic function $f(x) = x$ in the range $-\pi < x < \pi$.
- (b) Find the half range Fourier Sine series for $f(x) = \pi x - x^2$; $0 < x < \pi$ upto the first three terms.
- (c) Determine the Fourier series for the function $f(x) = x + x^2$ in the range

$$-\pi < x < \pi. \text{ Hence, show that } \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}.$$

Q 2

20 Marks [CO2]

- (a) Find the Laplace transform of the following function:

(i) $F(t) = e^{-t} \cos 3t$ (ii) $F(t) = t^3 e^{-t}$

- (b) Use Laplace transform method to solve the given initial value problem

$$\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + y(t) = \delta(t) ; \text{ given that } y(0) = 1 \text{ and } y'(0) = 1.$$

- (c) Find the Laplace transform of the function $F(t) = \frac{e^{at} - \cos bt}{t}$.

Q 3**20 Marks [CO3]**

- (a) Use the method of separation of variables to solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial y} + u; u(x, 0) = 6e^{-3x}$.
- (b) Use the method of separation of variables to solve $4 \frac{\partial u}{\partial y} = \frac{\partial u}{\partial x}; u(0, y) = 8e^{-3y}$.
- (c) Classify the following partial differential equations

- (i) $x \frac{\partial^2 z}{\partial x^2} - y \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} + \frac{\partial z}{\partial y} = 0$
- (ii) $2 \frac{\partial^2 u}{\partial x^2} - t \frac{\partial^2 u}{\partial x \partial t} + 2 \frac{\partial^2 u}{\partial t^2} + t \frac{\partial u}{\partial t} - \frac{\partial u}{\partial x} + u = 0$
- (iii) $\frac{\partial^2 u}{\partial t^2} + t \frac{\partial^2 u}{\partial x \partial t} + x \frac{\partial^2 u}{\partial x^2} + 2 \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} + u = 0$.

Q 4**20 Marks [CO4]**

- (a) Using Newton-Raphson method to compute a real positive root of the equation $x^3 - x - 1 = 0$ (Correct to three decimal places).
- (b) Using iteration method to find a real root of an equation $xe^x = 1$ correct to three significant digits.
- (c) Values of $f(x)$ in the interval $[0, 4]$ are given below:

x	0	1	2	3	4
f(x)	3	10	21	36	55

Using Simpson's one-third rule with a step size of 1, find the numerical approximation of $\int_0^4 f(x) dx$ (rounded off to the second decimal place).

Q 5**20 Marks [CO5]**

- (a) The number of airplane landings at an airport in a minute (X) and their probabilities are given by

X_i	0	1	2	3	4	5	6
$p(x_i)$	0.02	0.15	0.22	0.26	0.17	0.14	0.04

Find the mean and standard deviation of X.

(b) If the probability density function of a continuous random variable X is given by

$$f(x) = \begin{cases} kx^3; & 0 < x < 1 \\ 0; & \text{otherwise} \end{cases}$$

Find the value of k and hence compute its mean, variance, and standard deviation.

(c) A discrete random variable X has the probability function given below:

$X = x$	0	1	2	3	4	5	6	7	8
$P(x)$	0.01	0.03	0.05	0.07	0.09	0.11	0.13	0.15	0.17

Find the expected value and Standard deviation of random variable X .

